



Driver Drowsiness Detection System

Drowsy driving kills. Most systems are expensive or offline.

fatigue. We needed a low-cost, real-time system that works on a normal HP laptop with a loud alarm that actually wakes the driver.

FYP 203 • Miftah Uddin • Dir, KP

Real-time • Open Source



Problem: 20-30% crashes linked to fatigue

At night, on highways, drivers lose focus. Existing OEM systems are limited to expensive cars.



Need: Works on normal laptop

No GPU, no IR camera. Just HP webcam + Python. Must trigger loud siren instantly.



Motivation: Save lives locally

Open source on GitHub so students in Dir, Swat, Peshawar can replicate and improve.

Four clear goals for a working safety product.

Nota research prototype— a real alarm that beeps loudly when your eyes close, tested on HP.

✓ Verified: EAR 0.26 → 0.15 triggers instantly

01

Real-time eye tracking with EAR

MediaPipe Face Mesh 468 landmarks, p1- p6 calculation at 20 FPS on CPU

02

Loud alarm on drowsiness

winsound Beep 1000Hz + 1500Hz siren in separate thread, stops when awake

03

Works on normal HP laptop

find_camera() probes 0-4 + CAP_DSHOW fix for Windows
Unknown device error

04

GitHub open source

github.com/miftah9090/driver-drowsiness-detection-fyp with video demo 14.8 MB

Clean, documented structure that examiners love.

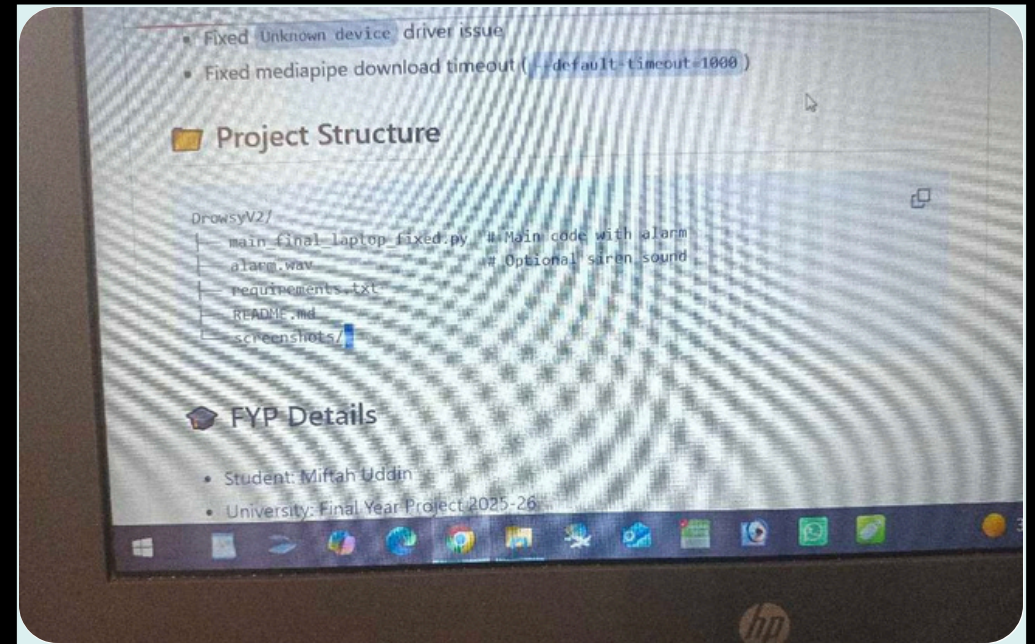
DrowsyV2/ — 5 files, working on HP after all fixes

DrowsyV2/

- main_final_laptop_fixed.py# Main code with alarm + CAP_DSHOW
- alarm.wav# Optional siren sound file
- requirements.txt
- README.md
- screenshots/
 - EAR_0.15_alert.jpg
 - demo.mp4 (14.8 MB)

✓ Fixed Unknown device driver issue — HP switch + find_camera() 0-4

✓ Fixed mediapipe download timeout (--default-timeout=1000)



FYP Details: Miftah Uddin • 2025-26

CMD: python main_final_laptop_fixed.py

q to quit, s to screenshot

EAR methodology that runs on CPU.

MediaPipe Face Mesh extracts 468 landmarks, we isolate 6 eye points per eye and compute blink closure ratio every frame.

$$\text{EAR} = (|p2-p6| + |p3-p5|) / (2 \times |p1-p4|)$$

DETECTION
468pts

FaceMeshlandmarks, refine=True, eyes only

THRESHOLD
0.25

<0.25=drowsy, >0.25 = awake

STABILITY
20
frames

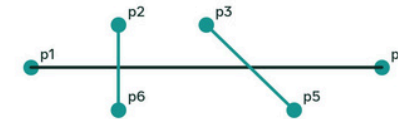
~1secclosedbefore alarm, avoids false blink

PERFORMANCE
~20
FPS

NoGPU,pureOpenCV + NumPy

Logic: (left_ear + right_ear)/2 → if <0.25 for 20 consecutive frames → trigger threading alarm → RED TEXT: DROWSINESS ALERT! BEEP BEEP! WAKE UP!

$$\text{EAR} = (|p2-p6| + |p3-p5|) / (2 \times |p1-p4|)$$



EAR > 0.25 = AWAKE | EAR < 0.25 = DROWSY



□ Camera probe 0-4 → Face Mesh 468 → EAR calc (p1-p6) → <0.25 & 20 frames → Alarm thread

FPS
20 FPS
on HP i5 CPU

LATENCY
< 1 sec
20 frames
threshold

Camera to alarm in 5 steps.

- 1 Find camera with Windows fix**
cv2.VideoCapture(i, CAP_DSHOW)probes0-4+phoneIP
fallback for Unknown device
- 2 MediaPipe Face Mesh 468 points**
Refine_landmarks=True tracks eyes even in dark, no dlib
model needed
- 3 $EAR = (p2-p6 + p3-p5) / 2*(p1-p4)$**
Left + right eye averaged every frame. Green dots drawn live.
- 4 Threshold check 0.25 + 20 frames**
If EAR < 0.25 for 20 consecutive frames → counter
triggers drowsiness
- 5 Alarm thread winsound siren**
1000Hz + 1500Hz BEEP in daemon thread non-blocking, stops
on eyes open

□ **KEY FIX FOR YOUR HP**
CAP_DSHOW prevents 0xA00F429E error. find_camera() tries
0,1,2,3,4 + IP Webcam URLs automatically.

☐ Debugged live in CMD • 3 fixes

Lenovo failed, HP worked. Real debugging.

Every error was logged, searched, and fixed with code.
External examiner can see problem-solving in GitHub commits and README.

Unknown device ✓

Timeout=1000 ✓

Alarm threading ✓



Lenovo 0xA00F429E — Something went wrong + Unknown device

Camera not opening on Lenovo, Device Manager shows Unknown device.

Fix → Switched to HP + find_camera() probes 0-4 + cv2.CAP_DSHOW + IP Webcam fallback



MediaPipe — Read timed out during install

pip install mediapipe fails on slow internet in Dir, KP.

Fix → pip install --default-timeout=1000 mediapipe==0.10.9 + protobuf==3.20.3



No alarm sound — detection works but silent

EAR drops to 0.15 but no wake-up effect.

Fix → winsound.Beep 1000Hz 500ms + 1500Hz 500ms siren loop in threading.Thread daemon

Working proof on HP laptop.

EAR drops from 0.26 AWAKE → 0.15 ALERT in ~1 sec (20 frames). Loud BEEP BEEP triggers instantly.

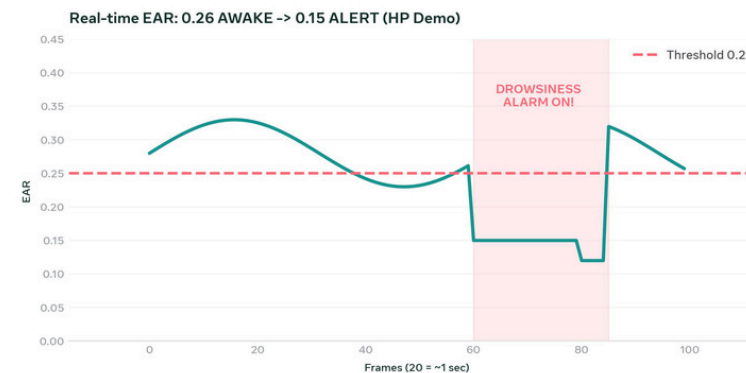


EAR^{0.15} → ALARM ON! BEEP BEEP! WAKE UP!

HP screenshot • FINAL FIXED • Green landmarks tracking

REAL-TIME EAR SIGNAL • HP DEMO

Threshold 0.25



0.26

AWAKE
eyes open

0.15

ALERT
eyes closed

20f

~1 sec
to alarm

fyp video.mp4 • 14.8 MB

CMD:pythonmain_final_laptop_fixed.py → q to quit, s to screenshot



Public repo • 5 files • 14.8 MB video

Open source, ready to run on HP.

driver-drowsiness-detection-fyp

LIVE • 1 commit

github.com/miftah9090/driver-drowsiness-detection-fyp

main_final_laptop_fixed.py

alarm.wav

requirements.txt

README.md

screenshots/

fyp video.mp4

INSTALLATION

```
# Fix Mediapipe timeout first
pip install --default-timeout=1000 -r requirements.txt

# requirements.txt
opencv-python
mediapipe==0.10.9
numpy • pygame • protobuf==3.20.3
```

RUN & DEMO

```
python main_final_laptop_fixed.py
# Auto finds camera: 0-4 + CAP_DSHOW + IP Webcam
# Close eyes → EAR 0.15 → ALARM ON BEEP BEEP!
```



Quit
Close window safely



Screenshot
Save JPG proof

□ FYP COMPLETE • HP VERIFIED • ALARM WORKING

Low-cost, real-time, working alarm system.

Tested on HP laptop after fixing Lenovo 0xA00F429E + Unknown device +

MediaPipe timeout. EAR 0.15 triggers loud 1000Hz + 1500Hz siren in threading — instantly wakes driver.



Yawning detection

MAR (Mouth Aspect Ratio) + EAR combo for drowsiness + fatigue



Head pose estimation

Pitch drop detection when driver nods off



IoT integration

Raspberry Pi + vibration motor in seat + GSM alert



Mobile app

Android app with IP Webcam + cloud logs for fleet tracking

Thank you — Questions?

□ github.com/miftah9090/driver-drowsiness-detection-fyp • Miftah Uddin • FYP 2023. Dir, KP